Chapter No 3:

Review Questions:

Question No 1:

Healthcare-as-a-Service:

Healthcare-as-a-Service delivers healthcare IT functions like electronic health records, telemedicine, medical data storage, and patient monitoring through cloud performs. Hospitals and clinics use these cloud services instead of building their own expensive infrastructure.

Haas helps improve patient care, provides remote consultation, and offers secure storage and sharing of medical data. it also reduce cost and enables faster access to advanced tools such as health analytics and AI-based diagnosis.

Education-as-a-Service (Eaas):

Education-as-a-Service provides learning tools, virtual classrooms, learning management systems (LMS) educational content and assessment platforms via cloud. School and Universities use

Eaas to support online classes, digital libraries collaboration, and remote learning.
it allows students and teachers to access educational resources anytime and anywhere.
Eaas reduces the needs for physical infrastructure and offers scalable, flexible, and interactive learning enviroments.

Question No 2 :

Classic Cloud Service Model	Hybrid Cloud Service Model
• Classic cloud Model uses only Public cloud services	• A Hybrid cloud Model use both Private and Public cloud
• All Applications, storage, and services run entirely on a cloud Provider like AWS, Azure	• Sensitive data or critical applications can stay inside the organization Private enviroment,
• it Provides high scalability and less maintaince work becuse it manages most Things.	• while less sensitive or high-demand services run on Public cloud.
• Classic is cheaper initially	• Hybrid offer better control over sesitive data.
• Classic for simple, cloud friendly app.	• Hybrid optimized cost by using Public control cloud for extra load.
	• Hybrid for enterprises with compliance needs

Question NO 3 :

Communication - as - a - Services (Caas) :

Communication - as - a service delivers communication tools like voice calling, video conferancing, messaging, chat, and SMS through the cloud. Instead of setting up their own communication system, organization subscribe to cloud - based communication services.

Caas makes it easy to integrate real-time communication into applications and business processes. It widely used for customer support centers, team collaboration, virtual, meetings and provides reliable communication without high infrastructure investment.

Question NO 4 :

Database - as - a - Service :

D'Baas provides database management through a cloud provider. Users can create, manage, and scale databases without handling hardware, installation, backups, or maintaince.

- The cloud Provider takes care of updates, security, replication, and automatic backups. Users simply access the database through APIs or dashboards.
- it provides high availability, easy scalling and reduced operational cost.

Question No 5:

Cloud Services are offered Amazon Web Services AWS:

AWS provides many cloud services including:

- Compelete Service: EC2 Lambda
- Storage services: S3 Glacier
- Databases: RDS DynamoDB
- Networking: VPC, Route 53
- Security tools: IAM, KMS
- Machine learning: Sage Maker
- Analytics: Redshift, Athena
- Developer tools: code Build, code Deploy
- IoT services and serverless computing features

These services allow organizations to build, deploy, and manage Application easily.

Question No 6:

Classic Cloud Service Model:

The classic Cloud Service Models include:

- **IaaS (Infrastructure-as-a Service):** Provides virtual Machine, storage and networking. Users control OS and application.
- **PaaS (Platform-as-a Service):** Provides a development platform including runtime, tools and databases. Developers only focus on apps, not infrastructure.
- **SaaS (Software-as-a-Service):** Ready-to-use application hosted on the cloud, accessed through a browser (e.g Gmail office 365).

These models offers different levels of control, flexibility and Management.

Question No 7:

Who are Service Providers? Classify:

Service Providers are companies that

Date:

deliver cloud computing services over the internet. They can be classified as:

1. Public Cloud Providers:

offer cloud services to the general public (AWS, Azure, Google cloud).

2. Private Cloud Providers:

Provide cloud services for a single organization.

3. Hybrid Cloud Providers:

Compute Engine, Big Query, Kubernetes Engine, cloud storage, AI/ML APIs.

4: Community Cloud Providers:

AI via waston, hybrid cloud tools, blockchain services. offer cloud infrastructure shared by organization with similar needs.

Question No 8:

Different Cloud Service Provider and Their Services:

compute (EC2), storage (S3) Databases (RDS), AI, IoT, Analytics

: Microsoft Azure:

Virtual Machines, SQL databases, Azure functions, Active Directory, Devops tools.

- **Google Cloud Plaform (GCP):**
Compute Engine, Big Query, Kubernetes Engine, cloud storage, AI/ML APIs.
- **IBM Cloud:**
AI via waston, hybrid cloud tools, blockchain services.
- **Oracle Cloud:**
Enterprise databases, cloud ERP, analytics compute and storage.

Question No 9:

Benefits of Network-as-a-Service (Naas):

- No Need to manage Physical network hardware
- Flexible and pay-as-you-go Pricing
- Fast deployment of network services.
- Managed security, Firewalls and VPN
- Easy scalling of bandwidth and
- Network capacity
- Centralized monitoring and management

Naas improves network performance while reducing cost and complexity.

Question No 10 :

List The different Service Provider for Cloud Computing :

1- Amazon Web Service (AWS) :

Largest cloud Provider. Offer computing, storage, analytics, AI, Networking, DevOPS, tools, and security services. Known for innovation and global data centers.

2- Microsoft Azure :

Strong in enterprise solutions. Offers virtual machines, SQL database, Active Directory, serverless computing and Microsoft 365 integrations

3- Google Cloud Platform :

Specializes in big data Analytics, AI, machine learning. Provides BigQuery, Kubernetes Engine and scalable compute service.

4- IBM Cloud :

Focuses on hybrid cloud, AI through version Waston, enterprise applications and secure cloud environments.

5- Oracle Cloud :

Best for Database-driven enterprises. Provides cloud ERP, big data tools high performance services